

Pre-feasibility summary

Crabbs Seawater Reverse-Osmosis Desalination Plant

Demand

Island demand estimated at ~34,000 m³/day by 2030; groundwater yield falls sharply in drought years.

Technology

SWRO chosen over thermal desalination: specific energy ~3.2 kWh/m³ versus 10+ for thermal.

Site

Geotechnical desk study favourable; intake depth ~12 m reached 450 m offshore.

Options

Three options compared: expand existing plant, new Crabbs plant, distributed containerised units. New plant ranked first on cost per m³ and resilience.

Conclusion

Proceed to full feasibility and procurement.